STUDENT ACADEMIC MILESTONE TRACKER

PROJECT REPORT

Submitted by

RAJ KUMAR R (7376231EC252)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

BANNARI AMMAN INSTITUTE OF TECHNOLOGY
(An Autonomous Institution Affiliated to Anna University ,Chennai)
SATHYAMANGALAM-638401

ANNA UNIVERSITY : CHENNAI 600025

OCTOBER-2025

BONAFIDE CERTIFICATE

Certified that this project report **"STUDENT ACADEMIC MILESTONR TRACKER"** is the bonafide work of **"RAJ KUMAR R (7376231EC252)"** who carried out the project work under my supervision.

Dr. Prakash S P

HEAD OF THE DEPARTMENT

Department of ECE

Bannari Amman Institute of Technology

Dr Sajan P Philip

ASSOCIATE PROFESSOR

Department of ECE

Bannari Amman Institute of Technology

Submitted for Project Viva Voce examination held on ……

Internal Examiner I

Internal Examiner II

DECLARATION

We affirm that the project work titled **"STUDENT ACADEMIC MILESTONE TRACKER**" being submitted in partial fulfillment for the award of the degree of **Bachelor of Engineering in Electronics and Communication Engineering is** the record of original work done by us under the guidance of **Dr. Sajan P Philip**, Associate Professor. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in

RAJ KUMAR R (7376231EC252)

I certify that the declaration made above by the candidates is true.

Dr.Sajan P Philip

Associate Professor

ACKNOWLEDGEMENT

We would like to enunciate heartfelt thanks to our esteemed Chairman **Dr. S.V.Balasubramaniam** and the respected Principal **Dr. C. Palanisamy** for providing excellent facilities and support during the course of study in this institute.

We are grateful to **Dr. Prakash S P, Head of the Department, Department of Electronics and Communication Engineering,** for his valuable suggestions to carry out the project work successfully.

We wish to express our sincere thanks to Faculty guide **Dr.Sajan p philip, Associate Professor, Department of Electronics and communication,** for his constructive ideas, inspirations, encouragement, excellent guidance, and needed technical support extended to complete our project work.

We would like to thank our friends, faculty and non-teaching staff who have directly and indirectly contributed to the success of this project.

RAJ KUMAR R (7376231EC252)

ABSTRACT

The Student Academic Milestone Tracker is a modern web-based platform designed to improve the way academic progress is monitored and managed. The system provides a centralized solution where students can easily view their academic milestones, including assignments, examinations, and project deadlines, through a simple and user-friendly interface. By offering a structured and organized view of academic activities, the platform helps students stay informed, manage their time effectively, and improve overall performance.

The system incorporates a secure authentication mechanism and role-based access control to ensure that only authorized users can access specific functionalities. Students are provided with a view-only portal to monitor their progress, while administrators have full control over managing student records, creating and updating milestones, and generating reports through a centralized dashboard. This ensures transparency, accuracy, and efficient academic management.

The platform is developed using modern technologies such as React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for database management, ensuring scalability, responsiveness, and efficient data handling. Real-time updates enable seamless communication between the frontend and backend, ensuring that all information is current and reliable.

Additionally, the system enhances usability by providing a clean and intuitive interface, reducing complexity and improving user experience. By replacing traditional manual tracking methods with an automated system, the Student Academic Milestone Tracker improves efficiency, minimizes errors, and ensures better academic monitoring. Overall, the platform serves as a secure, reliable, and scalable solution for modern educational institutions, supporting effective academic planning and management.

CHAPTER NO	TITLE	PAGE NO
1	INTRODUCTION	1
1.1	INTRODUCTION	1
1.2	PROBLEM STATEMENT	1
1.3	MOTIVATION	2
1.4	SCOPE OF THE PROJECT	2
1.5	OVERVIEW OF THE SYSTEM	3
1.6	OBJECTIVES	4
2	LITERATURE SURVEY	5
2.1	LITERATURE SURVEY	5
2.2	PRINCIPLES OF STUDENT ACADEMIC MANAGEMENT SYSTEMS	6
2.3	SECURE ACCESS AND DATA INTEGRATION	7
2.4	CENTRALIZED ADMIN DASHBOARDS	7
2.5	GAPS IDENTIFIED FROM LITERATURE	8
3	OBJECTIVES AND SYSTEM ARCHITECTURE	10
3.1	OBJECTIVES AND SYSTEM ARCHITECTURE	10
3.2	GOALS AND FEATURES	12
3.3	SYSTEM ARCHITECTURE	12
4	PROPOSED METHODOLOGY	16
4.1	METHODOLOGY	16
4.2	ANALYSIS OF REQUIREMENTS	16
4.3	FUNCTIONAL REQUIREMENTS	16
4.4	NON-FUNCTIONAL REQUIREMENTS	17
4.5	FEASIBILITY STUDY	17
4.6	EQUIPMENT AND TECHNOLOGY IDENTIFICATION	17
4.7	DEVELOPMENT OF MODULES	18
4.8	IMPLEMENTATION	20
4.9	VALIDATION AND TESTING	21
4.10	DEPLOYMENT, SERVER	23

4.11	MAINTENANCE AND ENHANCEMENT	24
4.12	USER INTERFACE (UI) AND USER EXPERIENCE (UX) DESIGN	26
4.13	ROLE-BASED ACCESS CONTROL (RBAC)	27
4.14	ERROR HANDLING AND VALIDATION MECHANISMS	27
4.15	SYSTEM PERFORMANCE MONITORING	27
5	RESULT AND DISCUSSION	29
5.1	RESULTS	29
5.2	MODULAR SYSTEM DESIGN	29
5.3	PERFORMANCE AND SECURITY	30
5.4	DISCUSSION	30
5.5	SUMMARY OF ACHIEVEMENTS	31
5.6	FUTURE SCOPE	32
6	CONCLUSION	35
6.1	CONCLUSION	35
	REFERENCES	37
	WORK CONTRIBUTION	39